

Hamza Mohammed Hassanain

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SUMMARY

Founding Software Engineer at **Repovive** (YC F'26), building the contest judge and core infrastructure for **10,000+ users**. **AWS Certified Solutions Architect**, **3× ICPC ACPC finalist**, and author of ICPC JCPC 2025 problem *G. Queries On Strings*. I contribute to **GitLab** and **LLVM** and write deep dives on dev.to, like [What Every Programmer Should Know About Memory](#).

EDUCATION

B.Sc. in Computer and Information Sciences — Suez Canal University

2022 – 2026

Graduation Project: [SigmaLoop](#), an AI tutor & curriculum engine (Grade: A+)

Ismailia, Egypt

Coursework: Data Structures & Algorithms, Operating Systems, Networks, Databases, Distributed Systems, Cloud Computing

EXPERIENCE

Repovive — YC F'26

Nov 2025 – Present

Founding Software Engineer (Full Time)

United States (Remote)

Repovive is a **Y Combinator-backed** science expert community where experts solve unsolved scientific problems, starting with theoretical computer science and math proof reasoning. We run **live programming and math contests**.

- Joined as the **first engineer** and built the core systems from the early days through **YC** and **10,000+ users**.
- Onboarded and supported new engineers, senior hires included, guiding them through the codebase and projects.
- Led design of the distributed judge (**judge1** on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by moving heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** test cases.
- Collaborated with the team to build AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build).
- Tracked down live-contest outages with **Claude Code**, **MCP servers**, **PostHog**, and log analysis, then shipped fixes.

OPEN SOURCE

GitLab

Nov 2025 – Jan 2026

Open Source Contributor

[MRs](#)

- Added typed group integrations to the Go SDK: Slack, Jira, MS Teams, Mattermost, Harbor, +3 [!2692](#), [!2675](#), [!2670](#)
- Added Harbor and Mattermost integration resources to the Terraform provider, plus a group SAML attribute [!2881](#), [!2909](#)
- Hardened **Json.safe_parse** and fixed Vue lint issues in the main GitLab codebase [!221163](#), [!221158](#)

LLVM

Nov 2025 – Feb 2026

Compiler Contributor

[PRs](#)

- Added constexpr evaluation of the X86 **cvtpd2ps** intrinsics in Clang [#169980](#)
- Lowered AArch64 and ARM CLS to **ISD::CTLs**, replacing a manual expansion [#178885](#), [#178430](#)
- Added LLDB SWIG Python extension tests for **SBTarget** [#173473](#)

ACHIEVEMENTS & CERTIFICATIONS

- **Certifications:** AWS Certified Solutions Architect – Associate (SAA-C03) [credly.com/verify](https://www.credly.com/verify)
- **Competitive Programming:** ICPC ACPC 3× Finalist (2024–2026); 17th of ~500 teams at ECPC 2024 icpc.global
- **Problem Author:** *G. Queries On Strings* (JCPC 2025), *F. Graph Tiling* (Repovive Round 3) codeforces.com | repovive.com

PROJECTS

Polyman CLI | *TypeScript, Node.js*

[npm](#) | [GitHub](#)

- Built a CLI that lets Codeforces problem setters create, test, and ship problems without leaving the terminal.
- One **verify** command runs generators, validators, checkers, and WA/TLE/RE solutions in C++, Java, and Python.
- Syncs with Polygon over its API (pull, push, commit, package); **1,600+ npm downloads** in the last year.

CP Judge on AWS | *Reference Architecture*

[GitHub](#)

- Designed Judge0 **isolate** on ECS-on-EC2 with a Fargate orchestrator and a Spot ASG scaled on queue depth.
- Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

SKILLS

Languages: C++, Go, Rust, Python, TypeScript, SQL | **Frameworks:** Node.js, Express.js, Next.js, React

Cloud & Tools: AWS, Docker, Nginx, Redis, Terraform, CI/CD, DigitalOcean, Linux | MongoDB, PostgreSQL

Concepts: System Design, Distributed Systems, Message Queues, REST APIs, WebSockets, Yjs (CRDTs)